Appendix C

Full Proofs

This appendix is the authoritative proof surface for the defended ORIUS theorem program, but it is not a flat headline ladder. The active audit distinguishes flagship defended rows, supporting defended rows, and visible draft/non-defended extensions. The battery witness remains the deepest proof chain, so the full step-by-step derivations below begin with the reference ladder:

1. OASG existence,
2. one-step safety preservation,
3. the ORIUS core bound,
4. no free safety,
5. certificate validity horizon,
6. certificate expiration bound,
7. feasible fallback existence, and
8. graceful degradation dominance.

The certificate half-life result remains a corollary, and safe-budget monotonicity remains a proposition rather than part of the T1–T8 reference ladder. The later extension theorems T9–T11 and L1–L4 are indexed here so that Chapter **??**, this appendix, and Appendix **??** point to the same proof surface, but those extension rows do *not* count toward the flagship defended novelty surface unless separately promoted by the active theorem audit. The broader integrated theorem-like surface, including precursor results and supporting chapter statements, is catalogued separately in Appendix **??**.

C.1 C.1 Battery-Domain OASG Existence

Theorem C.1 (restated)**.** *Under A1, A2, A4, A11, and A12, for dropout rate $d > 0$ and any optimal arbitrage controller, $\exists\, t^*$ with $\phi_{\text{obs}}(a_{t^*}) = 1$, $\phi_{\text{true}}(a_{t^*}) = 0$.*

Proof. Fix $\eta > 0$. By A11, the battery witness controller can be driven to a boundary-sensitive state under admissible demand, so there exists a step t^* and an admissible released action a_{t^*} such that the observed-state forward prediction lies on the safe side of the upper SOC boundary while remaining arbitrarily close to it:

$$\text{SOC}_{\max} - \eta \leq \hat{x}_{t^*} + \Delta(a_{t^*}) \leq \text{SOC}_{\max}.$$

By A2 and the standing dropout rate $d > 0$, there exists an admissible positive observation gap δ on a degraded-telemetry window. Choose that gap so that $\eta < \delta \leq g_{\max}$ and let

$$x_{t^*} = \hat{x}_{t^*} + \delta.$$

By A12, the fault process is conditionally independent of the controller action stream given the history, so the boundary-sensitive witness step and the admissible telemetry fault can co-occur on the same realized history. Now choose the admissible one-step model residual $\epsilon_{t^*} = 0$, which is allowed by A1 because $0 \in [-\epsilon_{\text{model}}, \epsilon_{\text{model}}]$.

The controller evaluates the action through the observed state only, so the observed successor satisfies

$$\hat{x}_{t^*+1} = \hat{x}_{t^*} + \Delta(a_{t^*}) \leq \text{SOC}_{\max},$$

hence $\phi_{\text{obs}}(a_{t^*}) = 1$. Using A4 for the shared one-step dynamics, the true successor is

$$x_{t^*+1} = x_{t^*} + \Delta(a_{t^*}) + \epsilon_{t^*} = \hat{x}_{t^*} + \Delta(a_{t^*}) + \delta.$$

Combining this identity with the lower side of the observed-state witness gives

$$x_{t^*+1} \geq (\text{SOC}_{\max} - \eta) + \delta = \text{SOC}_{\max} + (\delta - \eta) > \text{SOC}_{\max}.$$

Therefore $\phi_{\text{true}}(a_{t^*}) = 0$ while $\phi_{\text{obs}}(a_{t^*}) = 1$, which is exactly the battery witness-row observation-action safety gap. □

C.2 C.2 One-Step Safety Preservation

Theorem C.2 (restated)**.** *Under A1–A5, A7, if* $a_t^{\text{safe}} \in \mathcal{A}_t$ *and* $x_t \in \hat{C}_t^{\text{RAC}}$*, then* $\phi(x_{t+1}) = 1$*, where the tightened margin used to define* $\mathcal{A}_t$ *already absorbs the one-step model-error allowance.*

Proof. Fix a step t and assume the standing conditions A1–A5 and A7. Because $x_t \in \hat{C}_t^{\text{RAC}}$, the true state lies in the reliability-adjusted conformal tube issued from the current observation. In particular, membership in that tube gives

$$|x_t - \hat{x}_t| \leq m_t,$$

where m_t is the inflation-adjusted half-width determined by A2 and A5. This step is deterministic once the theorem is conditioned on $x_t \in \hat{C}_t^{\text{RAC}}$; it does not introduce a new probability statement. By A5, the shield uses the absorbed tightening margin

$$m_t^\star = m_t + \epsilon_{\text{model}}$$

when constructing $\mathcal{A}_t$. Since $a_t^{\text{safe}} \in \mathcal{A}_t$, the observed-state forward prediction therefore satisfies

$$\text{SOC}_{\min} + m_t^\star \leq \hat{x}_t + \Delta(a_t^{\text{safe}}) \leq \text{SOC}_{\max} - m_t^\star,$$

with $\Delta(a_t^{\text{safe}})$ denoting the one-step battery state update implied by the known dynamics from A4. Now write the true next state as

$$x_{t+1} = x_t + \Delta(a_t^{\text{safe}}) + \epsilon_t,$$

where the model residual ϵ_t is bounded by A1. Substituting $x_t = \hat{x}_t + (x_t - \hat{x}_t)$ and using $|x_t - \hat{x}_t| \leq m_t$ yields

$$x_{t+1} \geq (\hat{x}_t + \Delta(a_t^{\text{safe}})) - m_t - |\epsilon_t| \geq \text{SOC}_{\min} + m_t^\star - m_t - \epsilon_{\text{model}} = \text{SOC}_{\min} + (m_t + \epsilon_{\text{model}}) - m_t - \epsilon_{\text{model}} = \text{SOC}_{\min}.$$

The same substitution on the upper side gives

$$x_{t+1} \leq (\hat{x}_t + \Delta(a_t^{\text{safe}})) + m_t + |\epsilon_t| \leq \text{SOC}_{\max} - m_t^\star + m_t + \epsilon_{\text{model}} = \text{SOC}_{\max} - (m_t + \epsilon_{\text{model}}) + m_t + \epsilon_{\text{model}} = \text{SOC}_{\max}.$$

Therefore

$$\text{SOC}_{\min} \leq x_{t+1} \leq \text{SOC}_{\max}.$$

This is exactly the safety predicate $\phi(x_{t+1}) = 1$. The argument uses only the current-step

information certified by A7 and therefore preserves causality. Hence any action selected from $\mathcal{A}_t$ while the current state is inside $\hat{C}_t^{\mathrm{RAC}}$ keeps the one-step successor in the safe set. The theorem stops at this one-step conditional statement; it does not itself quantify how often $x_t \in \hat{C}_t^{\mathrm{RAC}}$ occurs. □

C.3 C.3 Battery ORIUS Core Bound

Theorem C.3 (restated)**.** *Under A1–A5, A7, and the explicit battery risk-envelope contract* $\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \leq \alpha(1 - w_t)$*, one has* $\mathbb{E}[V] \leq \alpha\, T\, (1 - \bar{w})$*.*

Proof. Let

$$V = \sum_{t=1}^{T} Z_t, \qquad Z_t = \mathbf{1}\Big[\neg\phi(x_t)\Big],$$

so that V counts the number of true-state safety violations across the episode. Because each Z_t is an indicator, $\mathbb{E}[Z_t] = \mathbb{P}[Z_t = 1]$. Applying linearity of expectation, which does not require independence across time, gives

$$\mathbb{E}[V] = \sum_{t=1}^{T} \mathbb{E}[Z_t] = \sum_{t=1}^{T} \mathbb{P}[Z_t = 1].$$

The theorem-local battery contract supplies the per-step conditional envelope

$$\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \leq \alpha(1 - w_t).$$

Taking expectations of both sides gives

$$\mathbb{P}[Z_t = 1] = \mathbb{E}[\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t]] \leq \alpha(1 - w_t).$$

Substituting the per-step bound into the expectation identity yields

$$\mathbb{E}[V] \leq \sum_{t=1}^{T} \alpha(1 - w_t) = \alpha \sum_{t=1}^{T} (1 - w_t).$$

Now expand the sum:

$$\sum_{t=1}^{T} (1 - w_t) = T - \sum_{t=1}^{T} w_t = T \left(1 - \frac{1}{T} \sum_{t=1}^{T} w_t\right).$$

By definition of the episode-average reliability,

$$\bar{w} = \frac{1}{T}\sum_{t=1}^{T} w_t.$$

Replacing the bracketed term with $\bar{w}$ gives

$$\mathbb{E}[V] \le \alpha\, T\,(1-\bar{w}).$$

This is the desired aggregation corollary. The actual burden is the per-step envelope hypothesis $\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le \alpha(1-w_t)$; once that T3a-style derivation is available, the displayed $\mathbb{E}[V] \le \alpha T(1-\bar{w})$ bound follows by linearity of expectation alone. The derivation invokes only finite-horizon identities and the explicit theorem-local battery risk budget; no temporal independence assumption is introduced. Therefore the battery ORIUS core bound holds for any admissible horizon T on the defended replay surface whenever that per-step budget contract has been justified. □

C.4 C.4 No Free Safety

Theorem C.4 (restated)**.** *Under A1, A2, A4, and A11, for any quality-ignorant π and $d > 0$, $\exists$ admissible fault sequence with* $\mathrm{TSVR}(\pi) > 0$.

Proof. Fix a quality-ignorant controller π. By definition, its effective safety margin does not depend on the runtime reliability score, so there exists a constant margin $m^\pi \ge 0$ used across all observation qualities. By A11, the battery witness controller can be driven to a boundary-sensitive state under admissible demand. Consequently, for every $\eta > 0$ there exists a step $t^\star$ and an action released by π such that the observed-state forward prediction nearly saturates the controller's fixed margin on the upper SOC boundary:

$$\hat{x}_{t^\star} + \Delta(\pi(\hat{x}_{t^\star})) \ge \mathrm{SOC}_{\max} - m^\pi - \eta.$$

Because $d > 0$, A2 admits an admissible degraded-observation gap. Choose a fault realization with

$$\delta \in (m^\pi + \eta,\ g_{\max}], \qquad x_{t^\star} = \hat{x}_{t^\star} + \delta.$$

Now set the admissible one-step model residual to zero, which is allowed by A1. Using A4 for the shared battery dynamics,

$$x_{t^\star+1} = x_{t^\star} + \Delta(\pi(\hat{x}_{t^\star})) + \epsilon_{t^\star} = \hat{x}_{t^\star} + \Delta(\pi(\hat{x}_{t^\star})) + \delta.$$

Substituting the observed-state witness inequality gives

$$x_{t^\star+1} \geq \mathrm{SOC}_{\max} - m^\pi - \eta + \delta = \mathrm{SOC}_{\max} + (\delta - m^\pi - \eta) > \mathrm{SOC}_{\max}.$$

The observed state therefore still appears feasible relative to the fixed margin m^π, while the same released action produces a true-state boundary crossing. Hence the controller incurs at least one true-state violation on the constructed admissible sequence, so $\mathrm{TSVR}(\pi) > 0$. The witness is finite-horizon and class-scoped exactly as claimed in T4. □

C.5 C.5 Certificate Validity Horizon

Theorem C.5 (restated)**.** *Under A1, A2, A4–A7, the battery safety certificate remains valid for the largest integer horizon τ_t whose forward tube stays inside the SOC constraint set. In the active audit this row is treated as a constructive definition rather than as part of the defended headline theorem count.*

Proof. The forward tube propagates the reliability-inflated interval under the certified safe action while adding the worst-case sub-Gaussian drift radius $\sigma_d\sqrt{\Delta}$. The certificate remains valid exactly while that tube is contained in $[\mathrm{SOC}_{\min}, \mathrm{SOC}_{\max}]$. The runtime helper `certificate_validity_horizon()` computes this maximal integer horizon directly from the chapter definition. Both T5 and T6 are registered in `THEOREM_REGISTER` (`src/orius/dc3s/theoretical_guarantees.py`) and enforced at runtime: the API router checks $\tau_t \geq 1$ at each step and fails the guarantee when violated. □

C.6 C.6 Certificate Expiration Bound

Theorem C.6 (restated)**.** *Under A4, A7, A9 and confidence level $\delta \in (0,1)$, the battery certificate horizon satisfies the lower bound*

$$\tau_t \geq \left\lfloor \frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \log(2/\delta)} \right\rfloor.$$

Proof. For a σ_d-sub-Gaussian disturbance, the standard tail inequality gives

$$\mathbb{P}\Big(|S_\Delta| \geq \delta_{\mathrm{bnd},t}\Big) \leq 2\exp\left(-\frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2\Delta}\right).$$

Demanding the right-hand side be at most δ and solving for Δ yields

$$\Delta \leq \frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \log(2/\delta)}.$$

Flooring gives the integer lower bound, and the runtime helper `certificate_expiration_bound()` implements this confidence-aware expression directly. □

C.7 C.7 Certificate Half-Life Corollary

Corollary C.7 (restated)**.** *The certificate half-life remains a corollary rather than part of the flagship T1–T8 ladder.*

Proof. The current battery blackout evidence supports a bounded safe-hold claim more strongly than a finite empirically identified half-life law. The corollary therefore remains analytical and should be interpreted together with the more conservative wording in Chapter **??**. □

C.8 C.8 Feasible Fallback Existence

Theorem C.8 (restated)**.** *Under A3, A4, A8, there exists a battery fallback action that preserves a safe SOC interior state under bounded model error.*

Proof. Zero dispatch is the constructive battery fallback. If the current SOC is in the interior of the safe interval and the one-step model error is bounded, then the next SOC remains inside the admissible interval. The helper `certify_fallback_existence()` checks this construction directly for the battery constraints. □

C.9 C.9 Graceful Degradation Dominance

Theorem C.9 (restated)**.** *Under A1–A5 and A8, for any paired graceful and uncontrolled violation sequences generated under the same admissible fault trace, the sequence-level comparison*

$$z_t^{\text{graceful}} \leq z_t^{\text{uncontrolled}} \quad \forall t$$

implies

$$V^{\text{graceful}} \leq V^{\text{uncontrolled}},$$

with strict inequality whenever the graceful sequence is strictly smaller on at least one step.

Proof. Summing the stepwise inequality over the shared trace gives

$$\sum_{t=1}^{T} z_t^{\text{graceful}} \leq \sum_{t=1}^{T} z_t^{\text{uncontrolled}},$$

which is exactly the displayed dominance statement for the cumulative violation counts. Strict inequality follows if the graceful sequence is strictly smaller on at least one step. The helper `evaluate_graceful_degradation_dominance()` implements this paired-sequence comparison directly. The appendix does not promote a broader absorbing-landing dynamics theorem beyond this scoped supporting surface. □

C.10 C.10 Safe-Budget Monotonicity Proposition

Proposition C.10 (restated)**.** *Under A1–A5, $B_t = |\mathcal{A}_t| \cdot \Delta t$ is monotone increasing in w_t.*

Proof. $|\mathcal{A}_t| = \mathrm{SOC}_{\max} - \mathrm{SOC}_{\min} - 2m_t$ and $m_t = q_t/(w_t + \epsilon)$ is monotone decreasing in w_t, so B_t increases with w_t. □

C.11 C.11 Universal Impossibility (T9)

Theorem reference. The authoritative statement is Theorem **??** in Chapter **??**. It is repeated here so that the full proof appendix covers the complete defended theorem program rather than only the battery witness ladder.

Appendix proof copy. By Assumption A10b, the fault process is φ-mixing with geometric decay, so a buffer gap of length τ_{mix} makes sufficiently separated windows approximately independent in total variation. Partition the horizon into analysis windows of length ℓ separated by those buffers. Inside each analysis window, invoke the battery no-free-safety witness from Section C.4: a controller that does not adapt its safety margin to the reliability process admits an admissible degraded-observation sequence with positive violation probability. The boundary-reachability and exogenous-fault structure are exactly the ingredients isolated in A11 and A12, so they package that window-level witness into a constant $c > 0$ such that $p_{\text{win}} \geq c\,d$ when the mean degradation rate is d.

Let Z_j^{win} indicate whether the jth separated window incurs a true-state violation and define

$$D_j = Z_j^{\text{win}} - \mathbb{E}\Big[Z_j^{\text{win}} \mid \mathcal{F}_{(j-1)(\ell+\tau_{\text{mix}})}\Big].$$

The sequence $\{D_j\}$ is a martingale difference sequence with $|D_j| \leq 1$. Azuma–Hoeffding therefore yields exponential concentration of the centered partial sums around their mean.

Since the expected number of bad windows is at least kcd over k separated windows, the probability that the total violation count drops below a constant fraction of kcd decays like $\exp(-\Omega(dT))$. Taking expectations gives $\mathbb{E}[V_T] \geq \Omega(dT)$, which is the claimed $\mathbb{E}[V_T] \geq c\,d\,T$ lower bound up to renaming the constant. □

C.12 C.12 Boundary-Indistinguishability Lower Bound (T10)

Theorem reference. The authoritative statement is Theorem **??** in Chapter **??**. The proof below preserves its scoped interpretation as a lower-bound frontier theorem under explicit extra hypotheses.

Appendix proof copy. Fix a step t and the binary boundary-testing subproblem with latent hypotheses $H_{0,t}$ and $H_{1,t}$. Because the controller only acts through the degraded observation channel, its decision is a binary test between the observation laws $P_{0,t}$ and $P_{1,t}$. By A13,

$$\mathrm{TV}(P_{0,t}, P_{1,t}) \leq w_t.$$

Therefore Le Cam's lemma gives

$$\max\Big(\mathbb{P}[\psi_t \text{ errs} \mid H_{0,t}], \mathbb{P}[\psi_t \text{ errs} \mid H_{1,t}]\Big) \geq \frac{1 - \mathrm{TV}(P_{0,t}, P_{1,t})}{2} \geq \frac{1 - w_t}{2}.$$

Whenever the unsafe-side hypothesis $H_{1,t}$ holds and the controller makes that binary mistake, the step contributes a true-state violation. Therefore

$$\mathbb{P}[Z_t = 1] \geq \mathbb{P}(H_{1,t})\,\mathbb{P}[\psi_t \text{ errs} \mid H_{1,t}] \geq p_t\,\frac{1 - w_t}{2}.$$

Summing over the horizon and applying linearity of expectation gives

$$\mathbb{E}[V_T(\pi)] \geq \frac{1}{2}\sum_{t=1}^{T} p_t(1 - w_t).$$

The special case $p_t \geq \alpha/2$ for all t yields the stated frontier $\mathbb{E}[V_T(\pi)] \geq \frac{1}{4}\alpha(1-\bar{w})T$. The proof does not claim a sharper constant without a domain-specific argument for the boundary-mass sequence $\{p_t\}$, which is exactly why T10 is defended as a stylized lower bound rather than as a global minimax theorem. □

C.13 C.13 Typed Structural Transfer and Failure-Mode Converse (T11)

Theorem reference. The forward theorem is Theorem **??** and the converse witness is Proposition **??** in Chapter **??**. The appendix copies the key proof logic so the full-proofs surface remains self-contained.

Appendix proof copy. Fix a step t and condition on the causal observation history $\mathcal{H}_t$. Let $E_t = \{x_t \in U_t\}$ denote the event that the latent state lies in the constructed observation-consistent set. By the first T11 obligation,

$$\mathbb{P}[E_t \mid \mathcal{H}_t] \geq 1 - \alpha.$$

On E_t, the soundness obligation implies that every action in the tightened safe-action set preserves safety at the next step for every state in U_t and every admissible disturbance. If the tightened safe-action set is non-empty, the repair-membership obligation guarantees that the released repaired action lies inside that set and is therefore safe on E_t. If the tightened safe-action set is empty, the fallback-admissibility obligation guarantees a safe release path from the current safe state. In either branch,

$$\mathbb{P}[x_{t+1} \in \mathcal{S} \mid \mathcal{H}_t] \geq \mathbb{P}[E_t \mid \mathcal{H}_t] \geq 1 - \alpha.$$

This proves the forward one-step transfer direction. The converse is not claimed as part of T11 itself; it is tracked separately as a structural failure-mode witness showing that if any one of the four obligations fails, the corresponding step in the forward proof disappears. □

C.14 L1 — Rate-Distortion Safety Law

Theorem C.11 (L1: Rate-Distortion Safety Law)**.** *Let $X \in \{0, 1\}$ denote the latent safe/unsafe state with entropy $H(X)$, let the observation channel have capacity C bits per use, and let $\alpha \in (0, 1)$ be the conformal miscoverage level. On the* stylized binary-source surrogate *used by the ORIUS law helper, we define the lower-envelope safety distortion by*

$$D^*(C) \;\geq\; \alpha \max\Big(0,\, 1 - C/H(X)\Big).$$

Proof. This law is a scoped modeling surrogate rather than a standalone information-theoretic converse. The ORIUS extension surface adopts the normalised unresolved-state fraction

$\max(0, 1 - C/H(X))$ as a binary-source distortion envelope and scales it by the runtime risk budget α. Substituting that surrogate definition yields the displayed lower envelope immediately. The clamp at zero handles the full-channel regime $C \geq H(X)$. □

Worked example. Battery SOC monitoring: $H(X) = 1$ bit (binary safe/unsafe), $C = 0.7$ bits (30% packet loss degrades the channel), $\alpha = 0.10$. Then $D^* \geq 0.10 \times (1 - 0.7) = 0.03$: at least a 3% safety-violation proxy is reported by the stylized law helper. This is a bounded design calculation, not a defended impossibility claim for the live battery pipeline.

Scope boundary / non-claim. L1 is kept in the appendix because it explains the extension-law calculators, but the active audit does *not* treat this linear lower envelope as a fully discharged Fano-tight converse.

C.15 L2 — Capacity Bridge

Theorem C.12 (L2: Capacity Bridge)**.** *Let $w_t \in [0, 1]$ be the OQE reliability weight. On a fixed evaluation surface where a domain-specific proxy constant $\kappa_d > 0$ has been calibrated, the ORIUS law helper uses the bridge*

$$\bar{w} \;\leq\; \kappa_d \frac{C}{H(X)},$$

as a capacity proxy.

Proof. This appendix now treats L2 as an explicit proxy bridge, not as a proved theorem of the live OQE. Once κ_d is fixed for the chosen surface, the displayed inequality is exactly the bridge relation that the code helper checks. A separate empirical or theoretical identification of κ_d is required outside this appendix. □

Worked example. With $C = 0.7$ bits, $H(X) = 1$ bit, $\kappa_d = 0.95$ (estimated on battery SCADA data), the bridge gives $\bar{w} \leq 0.95 \times 0.70 = 0.665$. An episode with $\bar{w} = 0.72$ would violate the bridge and signal miscalibration.

Scope boundary / non-claim. The active audit treats L2 as a domain-scoped proxy relation. It is not currently defended as a theorem deriving the heuristic OQE score from mutual information.

C.16 L3 — Critical Capacity Theorem

Theorem C.13 (L3: Critical Capacity)**.** *For any target violation rate $\varepsilon < \alpha$, define the stylized critical capacity*

$$C_d^* \;=\; \frac{H(X)\,(1-\varepsilon/\alpha)}{\kappa_d}.$$

Then any certification argument that uses the L2 proxy bridge together with the executable T3-style upper envelope requires $C \geq C_d^*$.

Proof. The executable upper envelope gives TSVR $\leq \alpha(1-\bar{w})$, so demanding TSVR $\leq \varepsilon$ forces $\bar{w} \geq 1-\varepsilon/\alpha$. Under the L2 proxy bridge, $\bar{w} \leq \kappa_d\, C/H(X)$, hence

$$1-\varepsilon/\alpha \;\leq\; \kappa_d\,\frac{C}{H(X)}.$$

Rearranging yields $C \geq C_d^*$. Thus C_d^* is the threshold required by the stylized bridge-and-achievability argument. □

Worked example. $\alpha = 0.10$, $\varepsilon = 0.02$, $\kappa_d = 1$, $H(X) = 1$ bit. $C_d^* = 1\times(1-0.02/0.10)/1 = 0.80$ bits. Within the stylized bridge argument, any channel with $C < 0.80$ bits fails to support a 2% TSVR certificate.

Scope boundary / non-claim. L3 is not presented as an independent universal impossibility theorem. It is an algebraic threshold derived from the L2 proxy and the T3-style upper envelope.

C.17 L4 — Achievability-Converse Sandwich

Theorem C.14 (L4: Achievability-Converse Sandwich)**.** *For the binary safe/unsafe state model:*

$$\frac{\alpha}{K}\,(1-\bar{w}) \;\leq\; \mathrm{TSVR}^* \;\leq\; \alpha\,(1-\bar{w}),$$

where the upper side is the executable T3-style envelope and the lower side is the stylized L1/L2 proxy lower envelope. The constant K captures the surrogate gap and does not grow with T.

Proof. **Upper side (executable achievability).** The defended T3-style envelope gives TSVR $\leq \alpha(1-\bar{w})$.

Lower side (stylized proxy). L1 contributes the surrogate lower envelope in capacity space and L2 maps that lower envelope into reliability space through the proxy constant κ_d.

Writing $K = 1/\kappa_d$ yields the displayed lower side

$$\frac{\alpha}{K}(1 - \bar{w}).$$

Combining the executable upper side with the stylized lower side gives the displayed sandwich. □

Worked example. $\bar{w} = 0.85$, $\alpha = 0.10$, $K = 2$. Lower: $(0.10/2)(1-0.85) = 0.0075$. Upper: $0.10 \times (1 - 0.85) = 0.015$. The stylized law family spans [0.75%, 1.5%]. This interval is a proxy comparison band, not a sharp theorem about the locked battery release.

Scope boundary / non-claim. The active audit treats L4 as an executable upper bound plus a stylized proxy lower side. It is *not* promoted as a closed optimality characterization.

C.18 T11_Byzantine — Byzantine-Tolerant OQE Bound

Theorem C.15 (T11_Byzantine: Byzantine Trimmed-Mean Bound)**.** *Given a window of W sensor readings with at most fraction $f < 1/3$ Byzantine (adversarial) sensors, the symmetric trimmed mean with trim fraction $\beta \geq f$ on each side and under the explicit* tail-contamination model *in which every Byzantine reading lies in one of the trimmed tails satisfies*

$$|\hat{\mu}_{\text{trim}} - \mu_{\text{true}}| \;\leq\; \frac{2\sigma_{\text{honest}}}{\sqrt{W(1-2f)}}$$

with probability $\geq 1 - 2e^{-2} \approx 0.729$.

Proof. Under the stated tail-contamination model, trimming βW readings from each tail removes all adversarial values because $\beta \geq f$ on each side. The remaining

$$W_{\text{eff}} = W(1-2f)$$

readings are therefore honest, and their dispersion is controlled by σ_{honest}.

By Hoeffding's inequality applied to W_{eff} i.i.d. sub-Gaussian readings with parameter σ_{honest}:

$$\Pr\Big[|\hat{\mu}_{\text{trim}} - \mu_{\text{true}}| > t\Big] \;\leq\; 2\exp\Big(-\frac{W_{\text{eff}}\, t^2}{2\sigma_{\text{honest}}^2}\Big).$$

Setting

$$t = \frac{2\sigma_{\text{honest}}}{\sqrt{W_{\text{eff}}}}$$

gives

$$\Pr\Big[|\hat{\mu}_{\text{trim}} - \mu_{\text{true}}| > t\Big] \le 2e^{-2}.$$

Hence the stated width holds with success probability at least $1-2e^{-2} \approx 0.729$. The threshold $f < 1/3$ ensures $W_{\text{eff}} > W/3 > 0$, so the trimmed window is non-empty. □

Worked example. $W = 30$ readings, $f = 0.10$ (3 adversarial), $\sigma_{\text{honest}} = 2.0\,\text{MW}$. $W_{\text{eff}} = 30 \times 0.80 = 24$. Bound: $4.0/\sqrt{24} = 0.816\,\text{MW}$. The trimmed mean is within 0.82 MW of the true value with $\ge 73\%$ confidence even under adversarial corruption.

C.19 T_stale_decay — Stale-Decay Graceful Degradation

Theorem C.16 (T_stale_decay: Stale-Decay Schedule)**.** *Under stale telemetry persisting for $k > \tau_{\max}$ steps (Assumption A6), the OQE reliability weight decays geometrically:*

$$w_t(k) \;=\; w_0\,\gamma^{k-\tau_{\max}}, \qquad k > \tau_{\max},$$

where $\gamma \in (0,1)$ is the decay rate. The weight reaches the floor ε at step

$$N_\varepsilon \;=\; \tau_{\max} + \left\lceil \frac{\ln(\varepsilon/w_0)}{\ln\gamma} \right\rceil.$$

Moreover, the episode TSVR under stale decay satisfies $\text{TSVR} \le \alpha\Big(1 - \bar{w}_{\text{stale}}\Big)$ *where $\bar{w}_{\text{stale}}$ accounts for the geometric series.*

Proof. During $k \le \tau_{\max}$, the telemetry is within the detection lag bound (A6) and $w_t = w_0$. For $k > \tau_{\max}$, each additional step of staleness multiplies the previous weight by γ: $w_t(k) = w_0\gamma^{k-\tau_{\max}}$ by induction.

Setting $w_0\gamma^{N-\tau_{\max}} = \varepsilon$ and solving: $N - \tau_{\max} = \ln(\varepsilon/w_0)/\ln\gamma$. Taking the ceiling gives N_ε.

The episode-average reliability over T steps with staleness onset at step t_0 is:

$$\bar{w} \;=\; \frac{1}{T}\Big[\sum_{k=1}^{\tau_{\max}} w_0 + \sum_{k=\tau_{\max}+1}^{T} w_0\gamma^{k-\tau_{\max}}\Big] \;=\; \frac{w_0}{T}\Big[\tau_{\max} + \frac{1-\gamma^{T-\tau_{\max}}}{1-\gamma}\Big].$$

The TSVR bound follows from T3 (Core Bound): $\text{TSVR} \le \alpha(1-\bar{w})$.

Remark. The geometric decay is a *design choice*, not a derived property of the physical system. It encodes the principle that stale observations carry exponentially less information,

consistent with the channel-capacity interpretation (capacity of an erasure channel with increasing erasure probability). □

Worked example. $w_0 = 0.95$, $\gamma = 0.85$, $\tau_{\max} = 6$ steps, $\varepsilon = 0.05$. $N_\varepsilon = 6 + \lceil \ln(0.05/0.95)/\ln(0.85) \rceil = 6 + \lceil -2.944/(-0.1625) \rceil = 6 + 19 = 25$ steps. After 25 steps of persistent staleness the system enters safe-landing mode ($w_t \leq 0.05$).

C.20 T_minimax — Tight OASG Minimax Tradeoff

Theorem C.17 (T_minimax: Minimax OASG Tradeoff)**.** *Under degraded observation with mean reliability $\bar{w}$, the current extension surface combines a stylized lower envelope with the defended T3-style upper envelope:*

$$\frac{\alpha}{K}\sum_{t=1}^{T}(1-w_t) \;\leq\; \mathbb{E}[V_T] \;\leq\; \alpha\sum_{t=1}^{T}(1-w_t),$$

where $K = 2$ for binary state spaces and $V_T = \sum_{t=1}^{T} Z_t$ is the cumulative violation count.

Proof. **Upper bound.** By T3 (Core Bound), DC3S ensures $\Pr[Z_t = 1 \mid \mathcal{H}_t] \leq \alpha(1 - w_t)$, so linearity of expectation gives $\mathbb{E}[V_T] \leq \alpha \sum_t (1 - w_t)$.

Lower bound. The extension surface adopts the stylized L1/L2 proxy lower envelope $r_t \geq (\alpha/K)(1 - w_t)$. This follows because:

1. L1 gives $D^*(C_t) \geq \alpha(1 - C_t/H(X))$ where C_t is the effective per-step channel capacity.
2. L2 translates: $1 - w_t \leq 1 - \kappa_d C_t/H(X)$, i.e., $1 - C_t/H(X) \geq (1 - w_t)/\kappa_d$.
3. Substituting: $r_t \geq \alpha(1 - w_t)/\kappa_d$. With $K = 1/\kappa_d$ (which equals 2 for BSC at worst-case crossover): $r_t \geq (\alpha/K)(1 - w_t)$.

Summing over T steps and taking expectations completes the proxy sandwich. □

Worked example. $T = 100$ steps, $\bar{w} = 0.90$, $\alpha = 0.10$, $K = 2$. Lower: $(0.10/2) \times 100 \times 0.10 = 0.50$ expected violations. Upper: $0.10 \times 100 \times 0.10 = 1.00$ expected violations. The resulting interval is a stylized comparison band. It is not a sharp minimax certificate for the locked deployment artifacts.

Scope boundary / non-claim. $\mathrm{T}_{\mathrm{minimax}}$ remains a scoped extension theorem because its lower side inherits the open L1/L2 proxy bridge.

C.21 T_sensor_converse — Sensor Quality Converse

Theorem C.18 (T_sensor_converse: Sensor Quality Converse)**.** *For any controller π and target* $\mathrm{TSVR}(\pi) \le \varepsilon$*:*

$$\bar{w} \;\ge\; 1 - \frac{\varepsilon}{\alpha}.$$

This threshold is sufficient by the executable T3-style envelope and necessary only inside the stylized L2/L3 bridge model.

Proof. **Stylized necessity.** Under the L2/L3 proxy bridge, $\bar{w} < 1 - \varepsilon/\alpha$ means the stylized threshold is not met.

Executable sufficiency. If $\bar{w} \ge 1 - \varepsilon/\alpha$, then $\alpha(1 - \bar{w}) \le \varepsilon$, and DC3S achieves $\mathrm{TSVR} \le \alpha(1 - \bar{w}) \le \varepsilon$. □

Worked example. $\alpha = 0.10$, $\varepsilon = 0.02$. $\bar{w} \ge 1 - 0.02/0.10 = 0.80$. To guarantee $\le 2\%$ violations, mean sensor reliability must be at least 80% according to the stylized threshold calculator.

Scope boundary / non-claim. The active audit does not treat $\mathrm{T}_{\mathrm{sensor}}$ as a fully defended universal converse for all controllers; it is the inverse of the executable upper envelope plus the scoped L2/L3 bridge.

C.22 T_trajectory_PAC — Distribution-Free Trajectory Safety Certificate

Theorem C.19 (T_trajectory_PAC: PAC Trajectory Certificate)**.** *Under A1, A4, A5, A9, and the implemented Bonferroni/union-bound aggregation over the horizon, for horizon H, calibration size n_{cal}, and confidence target $1 - \delta$:*

$$\Pr\Big[\textit{all } H \textit{ steps safe}\Big] \;\ge\; 1 - H\alpha(1 - \bar{w}) - \varepsilon_{\mathrm{fs}} - \delta/2,$$

where the implemented finite-sample correction is

$$\varepsilon_{\mathrm{fs}} = \sqrt{\frac{\ln(4/\delta)}{2n_{\mathrm{eff}}}}, \qquad n_{\mathrm{eff}} = \Big\lfloor n_{\mathrm{cal}} \min_{t \le H} w_t \Big\rfloor.$$

The reliability-deflated n_{eff} is a conservative executable calibration rule used by the helper, not a claim of optimal split-conformal sample complexity.

The maximum trajectory horizon certified at confidence $1-\delta$ by this bound is

$$H_{\max} = \max\left\{0,\ \left\lfloor \frac{\delta/2 - \varepsilon_{\text{fs}}}{\alpha(1-\bar{w})} \right\rfloor\right\}.$$

Proof. **Step 1 (per-step risk).** By T3, $\Pr[Z_t = 1 \mid \mathcal{H}_t] \le \alpha(1 - w_t)$.

Step 2 (trajectory aggregation by union bound). Let $E_t = \{Z_t = 1\}$ denote a stepwise true-state violation. Then

$$\Pr\left[\bigcup_{t=1}^{H} E_t\right] \ \le\ \sum_{t=1}^{H} \Pr[E_t] \ \le\ \sum_{t=1}^{H} \alpha(1-w_t) \ \le\ H\alpha(1-\bar{w}).$$

Step 3 (finite-sample correction). The conformal quantile $\hat{q}$ estimated from n_{cal} points satisfies (Vovk, 2005): $\Pr[\hat{q} < q_{\text{true}}] \le \varepsilon_{\text{fs}}$. The executable helper conservatively deflates the effective calibration size by the worst reliability over the certified prefix:

$$n_{\text{eff}} = \left\lfloor n_{\text{cal}} \min_{t \le H} w_t \right\rfloor.$$

Step 4 (combining).

$$\begin{aligned}
&\Pr[\exists\, t \in [H] : \text{violation}] \\
&\quad \le \underbrace{H\alpha(1-\bar{w})}_{\text{risk budget}} + \underbrace{\varepsilon_{\text{fs}}}_{\text{calibration error}} + \underbrace{\delta/2}_{\text{exit-time budget}} \\
&\Rightarrow\ \Pr[\text{all safe}] \ge 1 - H\alpha(1-\bar{w}) - \varepsilon_{\text{fs}} - \delta/2.
\end{aligned}$$

To certify the trajectory at confidence $1-\delta$, it is sufficient that

$$1 - H\alpha(1-\bar{w}) - \varepsilon_{\text{fs}} - \delta/2 \ge 1 - \delta.$$

Rearranging gives

$$H\alpha(1-\bar{w}) \le \delta/2 - \varepsilon_{\text{fs}}.$$

If the right-hand side is negative, the bound certifies no positive horizon. Otherwise flooring the quotient gives the displayed $H_{\max}$. □

Worked example. $\bar{w} = 0.90$, $\alpha = 0.10$, $n_{\text{cal}} = 300$, $\delta = 0.05$, $\min_t w_t = 0.60$. $n_{\text{eff}} = \lfloor 300 \times 0.60 \rfloor = 180$. $\varepsilon_{\text{fs}} = \sqrt{\ln(80)/(360)} = \sqrt{4.382/360} = 0.110$. Because $\delta/2 - \varepsilon_{\text{fs}} =$

$0.025 - 0.110 = -0.085 < 0$, the certificate is vacuous at the 95% confidence target and

$$H_{\max} = 0.$$

With this calibration size and reliability floor, the implemented helper honestly reports that no positive trajectory horizon is certified at the requested confidence level.

Scope boundary / non-claim. The defended executable surface for $\mathrm{T}_{\mathrm{PAC}}$ is the Bonferroni/union-bound certificate above. No separate Ville/supermartingale strengthening is claimed in the live repo.